Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The solution of the differential equation $\frac{dy}{dx} = \frac{1-3y-3x}{1+x+y}$ is

(A) $x + y - \ln|x + y| = c$

(B) $3x + y + 2\ell n|1 - x - y| = c$

(C) $x + 3y - 2\ln|1 - x - y| = c$

(D) None of these

Q.2

The solution of differential equation $yy' = x\left(\frac{y^2}{x^2} + \frac{f\left(y^2/x^2\right)}{f'\left(y^2/x^2\right)}\right)$ is

(A) $f\left(y^2/x^2\right) = cx^2$

(B) $x^2 f\left(y^2/x^2\right) = c^2y^2$

(C) $x^2 f\left(y^2/x^2\right) = c$

(D) $f\left(y^2/x^2\right) = cy/x$

Q.3

The expression which is the general solution of the differential equation $\frac{dy}{dx} + \frac{x}{1-x^2}y = x\sqrt{y}$ is

(A) $\sqrt{y} + \frac{1}{3}\left(1 - x^2\right) = c\left(1 - x^2\right)^{\frac{1}{4}}$

(B) $y\left(1 - x^2\right)^{\frac{1}{4}} = c\left(1 - x^2\right)$

(C) $\sqrt{y}\left(1 - x^2\right)^{\frac{1}{4}} = \frac{1}{3}\left(1 - x^2\right) + c$

(D) None of these

Q.4

The general solution of the differential equation $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1 + x)e^x \sec y$ is

(A) $\sin(1 + x) = y\left(e^x + c\right)$

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(B) $y\sin(1+x)=ce^x$

(C) $(1+x)\sin y=e^x+c$

(D) $\sin y=(1+x)(e^x+c)$

Q.5

Which one of the following differential equations represents the family of straight lines which are at unit distance from the origin?

(A) $\left(y-x\frac{dy}{dx}\right)^2=1-\left(\frac{dy}{dx}\right)^2$

(B) $\left(y+x\frac{dy}{dx}\right)^2=1+\left(\frac{dy}{dx}\right)^2$

(C) $\left(y-x\frac{dy}{dx}\right)^2=1+\left(\frac{dy}{dx}\right)^2$

(D) $\left(y+x\frac{dy}{dx}\right)^2=1-\left(\frac{dy}{dx}\right)^2$

Q.6

What is the solution of the differential equation
$a\left(x\frac{dy}{dx}+2y\right)=xy\frac{dy}{dx}$?

(A) $x^2=kye^{\frac{y}{a}}$

(B) $yx^2=kye^{\frac{y}{a}}$

(C) $y^2x^2=kye^{\frac{y^2}{a}}$

(D) None of the above

Q.7

The solution to the differential equation
$\frac{dy}{dx}=\frac{yf'(x)-y^2}{f(x)}$
where $\mathrm{f(x)}$ is a given function is

(A) $\mathrm{f(x)=y(x+c)}$

(B) $\mathrm{f(x)=cxy}$

(C) $f(x)=c(x+y)$

(D) $yf(x)=cx$

Q.8

The gradient of the curve passing through $(4,0)$ is

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given by $\frac{dy}{dx} - \frac{y}{x} + \frac{5x}{(x+2)(x-3)} = 0$ if the point $(5, a)$ lies on the curve, then the value of a is

(A) $\frac{67}{12}$

(B) $5 \sin \frac{7}{12}$

(C) $5 \log \frac{7}{12}$

(D) None of these

Q.9

The solution of the differential equation
$x \sin x \frac{dy}{dx} + (x \cos x + \sin x)y = \sin x$
When $y(0) = 0$ is

(A) $xy \sin x = 1 - \cos x$

(B) $xy \sin x + \cos x = 0$

(C) $x \sin x + y \cos x = 0$

(D) $x \sin x + y \cos x = 1$

Q.10

The solution to of the differential equation
$(x+1)\frac{dy}{dx} - y = e^{3x}(x+1)^2$ is

(A) $y = (x+1)e^{3x} + c$

(B) $3y = (x+1) + e^{3x} + c$

(C) $\frac{3y}{x+1} = e^{3x} + c$

(D) $ye^{-3x} = 3(x+1) + c$

Q.11

The solution of the differential equation
$\frac{dy}{dx} + \frac{2yx}{1+x^2} = \frac{1}{(1+x^2)^2}$ is

(A) $y\left(1+x^2\right) = c + \tan^{-1} x$

(B) $\frac{y}{1+x^2} = c + \tan^{-1} x$

(C) $y \log\left(1+x^2\right) = c + \tan^{-1} x$

(D) $y\left(1+x^2\right) = c + \sin^{-1} x$

Q.12

An integrating factor of the differential equation $\sin x \frac{dy}{dx} + 2y \cos x = 1$ is

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(A) $\sin^2 x$

(B) $\frac{2}{\sin x}$

(C) $\log|\sin x|$

(D) $\frac{1}{\sin^2 x}$

Q.13

The general solution the differential equation $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$ is

(A) $\sin(1+x) = y\left(e^x + c\right)$

(B) $y\sin(1+x) = ce^x$

(C) $(1+x)\sin y = e^x + c$

(D) $\sin y = (1+x)\left(e^x + c\right)$

Q.14

The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x}\log y = \frac{y}{x^2}(\log y)^2$ is

(A) $y = \log\left(x^2 + cx\right)$

(B) $\log y = x\left(cx^2 + \frac{1}{2}\right)$

(C) $x = \log y\left(cx^2 + \frac{1}{2}\right)$

(D) None of these.

Q.15

Solution of the differential equation $\frac{dx}{dy} - \frac{x\log x}{1+\log x} = \frac{e^y}{1+\log x}$, ify $(1) = 0$, is

(A) $x^x = e^{ye^y}$

(B) $\mathrm{e^y = x^{e^y}}$

(C) $x^x = ye^y$

(D) None of these

Q.16

The solution to the differential equation $\frac{dy}{dx} = \frac{yf'(x) - y^2}{f(x)}$ where $f(x)$ is a given function is

(A) $f(x) = y(x+c)$

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(B) $f(x) = cxy$

(C) $f(x) = c(x + y)$

(D) $yf(x) = cx$

Q.17

If $\phi(x)$ is a differentiable function, then the solution of the differential equation $dy + \{y\phi'(x) - \phi(x)\phi'(x)\}\, dx = 0$ is

(A) $y = \{\phi(x) - 1\} + ce^{-\phi(x)}$

(B) $y\phi(x) = \{\phi(x)\}^2 + c$

(C) $ye^{\phi(x)} = \phi(x)\mathrm{e}^{\phi(x)} + \mathrm{c}$

(D) None of these

Q.18

What is the solution of the differential equation $\frac{dx}{dy} + \frac{x}{y} - y^2 = 0$?

(A) $xy = x^4 + c$

(B) $xy = y^4 + c$

(C) $4xy = y^4 + c$

(D) $3xy = y^3 + c$

where c is an arbitrary constant.

Q.19

What is the solution of the differential equation $(x + y)(dx - dy) = dx + dy$?

(A) $x + y + \ln(x + y) = c$

(B) $x - y + \ln(x + y) = c$

(C) $y - x + \ln(x + y) = c$

(D) $y - x - \ln(x - y) = c$

Q.20

The curve satisfying the differential equation $\left(1 - x^2\right)\mathrm{y}' + x\mathrm{y} = \mathrm{a}x$ are

(A) ellipses and hyperbolas

(B) ellipses and parabola

(C) ellipses and straight lines

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(D) circles and ellipses

Answers & Solutions

Q.1 Answer:

$3x + y + 2\ell n|1 - x - y| = c$

Solution:

This is the form in which $\frac{a_1}{a_2} = \frac{b_1}{b_2}$ **The given equation can be rewritten as** $\frac{dy}{dx} = \frac{1-3(x+y)}{1+(x+y)} = f(x+y)$

Substitute $x + y = z \Rightarrow 1 + \frac{dy}{dx} = \frac{dz}{dx}$**. The equation then becomes.** $\frac{dz}{dx} - 1 = \frac{1-3z}{1+z} \Rightarrow \frac{dz}{dx} = \frac{1-3z+1+z}{1+z} = \frac{2-2z}{1+z}$

$\Rightarrow \frac{1+z}{2(1-z)}dz = dx$

On integrating we get $\frac{1}{2}\int \frac{1+z}{1-z}dz = \int dx + a \Rightarrow \frac{1}{2}\int \left[\frac{2}{1-z} - 1\right] dz = x + a$

$\Rightarrow -\ell n|1 - z| - \frac{1}{2}z = x + a$

$\Rightarrow -\ell n|1 - x - y| - \frac{1}{2}(x + y) = x + a$

$\Rightarrow -2\ell n|1 - x - y| - 3x - y = 2a$

$\Rightarrow 3x + y + 2\ell n|1 - x - y| = c$ **where** $c = -2a$

Q.2 Answer:

$f\left(y^2/x^2\right) = cx^2$

Solution:

The given equation can be written as $\frac{y}{x}\frac{dy}{dx} = \left\{\frac{y^2}{x^2} + \frac{f\left(y^2/x^2\right)}{f'(y^2/x^2)}\right\}$

The above equation is a homogeneous equation. Putting $y = vx$**, we get** $\left[v + x\frac{dv}{dx}\right] = v^2 + \frac{f\left(v^2\right)}{f'(v^2)}$

or $vx\frac{dv}{dx} = \frac{f\left(v^2\right)}{f'(v^2)}$ **(variable separable)**

or $\frac{2vf'\left(v^2\right)}{f(v^2)}dv = 2\frac{dx}{x}$

Now, integrating both sides, we get $\log f\left(v^2\right) = \log x^2 + \log c$[$\log c =$ **constant**]

or $\log f\left(v^2\right) = \log cx^2$ **or** $f\left(v^2\right) = cx^2$ **or** $f\left(y^2/x^2\right) = cx^2$

Q.3 Answer:

$\sqrt{y} + \frac{1}{3}\left(1 - x^2\right) = c\left(1 - x^2\right)^{\frac{1}{4}}$

Solution:

Divide the equation by $\sqrt{y}$**, we get** $y^{-\frac{1}{2}}\frac{dy}{dx} + \frac{x}{1-x^2}y^{\frac{1}{2}} = x$

Put $y^{\frac{1}{2}} = z \Rightarrow \frac{1}{2}y^{-\frac{1}{2}}\frac{dy}{dx} = \frac{dz}{dx}$

$2\frac{dz}{dx} + \frac{x}{1-x^2}z = x \Rightarrow \frac{dz}{dx} + \left(\frac{1}{2}\frac{x}{1-x^2}\right)z = \frac{x}{2}$

I.F. $= e^{\int \frac{1}{2}\left[\frac{x}{1-x^2}\right]dx} = e^{-\frac{1}{4}\log\left(1-x^2\right)} = \left(1 - x^2\right)^{-\frac{1}{4}}$

The solution $z\left(1 - x^2\right)^{-\frac{1}{4}} = \int \frac{x}{2}(1 - x)^{-\frac{1}{4}}dx + c$

$\Rightarrow z\left(1 - x^2\right)^{-\frac{1}{4}} = \frac{1}{2}\int \left(1 - x^2\right)^{-\frac{1}{4}} xdx + c$

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$\Rightarrow \sqrt{y}\left(1-x^2\right)^{-\frac{1}{4}}=\frac{1}{2}\left(-\frac{1}{2}\right)\frac{\left(1-x^2\right)^{3/4}}{3/4}+c$

$\Rightarrow \sqrt{y}\left(1-x^2\right)^{-\frac{1}{4}}=-\frac{1}{3}\left(1-x^2\right)^{3/4}+c$

$\Rightarrow \sqrt{y}+\frac{1}{3}\left(1-x^2\right)=c\left(1-x^2\right)^{\frac{1}{4}}$

Q.4 Answer:

$\sin y=(1+x)\left(e^x+c\right)$

Solution:

Divide the equation by sec y $\cos y\frac{dy}{dx}-\frac{\sin y}{1+x}=(1+x)e^x$

Put $\sin y=z\Rightarrow \cos y\frac{dy}{dx}=\frac{dz}{dx}$ **then**

$\frac{dz}{dx}-\left(\frac{1}{1+x}\right)z=(1+x)e^x$:

I.F. $=e^{-\int\frac{1}{1+x}dx}=e^{-\log(1+x)}=\frac{1}{1+x}$

The solution is $z\left(\frac{1}{1+x}\right)=e^x+c\Rightarrow \sin y=(1+x)\left(e^x+c\right)$

Q.5 Answer:

$\left(y-x\frac{dy}{dx}\right)^2=1+\left(\frac{dy}{dx}\right)^2$

Solution:

$y=mx+c$ **(Equation of straight line)** $\frac{dy}{dx}=m$ **and** $mx-y+c=0$ **is at unit distance from origin.** $\therefore \frac{|m(0)-(0)+c|}{\sqrt{m^2+(-1)^2}}=1\Rightarrow c=\sqrt{1+m^2}$

Now:

$\left[y-x\frac{dy}{dx}\right]^2=[mx+c-xm]^2=c^2=1+m^2$

also,

$\left[y+x\frac{dy}{dx}\right]^2=[mx+c+mx]^2=\left[2mx+\sqrt{1+m^2}\right]^2$

also, $1-\left(\frac{dy}{dx}\right)^2=1-m^2$ **and** $1+\left(\frac{dy}{dx}\right)^2=1+m^2$

$\Rightarrow \left[y-x\frac{dy}{dx}\right]^2=1+\left(\frac{dy}{dx}\right)^2$

Q.6 Answer:

None of the above

Solution:

Given differential equation is $a\left(x\frac{dy}{dx}+2y\right)=xy\frac{dy}{dx}$

$\Rightarrow \quad ax\frac{dy}{dx}-xy\frac{dy}{dx}=-2ay$

$\Rightarrow \quad (xy-ax)\frac{dy}{dx}=2ay\Rightarrow x(y-a)\frac{dy}{dx}=2ay$

$\Rightarrow x(y-a)dy=2aydx$

$\Rightarrow \frac{(y-a)}{y}dy=\frac{2a}{x}dx\Rightarrow\left(1-\frac{a}{y}\right)dy=\frac{2a}{x}dx$

$dy-\frac{a}{y}dy=\frac{2a}{x}dx$

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Integrate on both side $\int dy - a\int \frac{1}{y}dy = 2a\int \frac{1}{x}dx$

$y - a\log y = 2a\log x + \log c$

$\Rightarrow y = a\log x^2yc \Rightarrow x^2y = ke^{y/a}$

Q.7 Answer:

$\mathrm{f(x) = y(x + c)}$

Solution:

We have $\frac{dy}{dx} = \frac{f'(x)}{f(x)}y - \frac{y^2}{f(x)}$

$\Rightarrow \frac{dy}{dx} - \frac{f'(x)}{f(x)}y = -\frac{y^2}{f(x)}$

Divide by $y^2 y^{-2}\frac{dy}{dx} - y^{-1}\frac{f'(x)}{f(x)} = -\frac{1}{f(x)}$

Put $y^{-1} = z \Rightarrow -y^{-2}\frac{dy}{dx} = \frac{dz}{dx}$

$-\frac{\mathrm{dz}}{\mathrm{dx}} - \frac{\mathrm{f'(x)}}{\mathrm{f(x)}}(\mathrm{z}) = -\frac{1}{\mathrm{f(x)}}$

$\Rightarrow \frac{\mathrm{dz}}{\mathrm{dx}} + \frac{\mathrm{f'(x)}}{\mathrm{f(x)}}\mathrm{z} = \frac{1}{\mathrm{f(x)}}$

I.F. $= e^{\int \frac{f'(x)}{f(x)}dx} = e^{\log f(x)} = f(x)$

∴ The solution is $z(f(x)) = \int \frac{1}{f(x)}(f(x))dx + c$

$\Rightarrow \mathrm{y^{-1}(f(x)) = x + c \Rightarrow f(x) = y(x + c)}$

Q.8 Answer:

$5\log\frac{7}{12}$

Solution:

The differential equation is $\frac{dy}{dx} - \frac{y}{x} = -\frac{5x}{(x+2)(x-3)}$

I. $F = e^{\int\left(-\frac{1}{x}\right)dx} = e^{-\ln x} = \frac{1}{x}$

Solution is

$\mathrm{y\left(\frac{1}{x}\right) = \int\left(\frac{1}{x}\right) \times \frac{5x}{(x+2)(x-3)}dx = \ln\left(\frac{x+2}{x-3}\right) + C}$

It passes through $(4, 0)$**, so** $\vec{C} = -\ln 6 \therefore \mathrm{y = x\ln\left\{\frac{(x+2)}{6(x-3)}\right\}}$ **Putting** $(5, a)$

we get $a = 5\ln\left(\frac{7}{12}\right)$

Q.9 Answer:

$xy\sin x = 1 - \cos x$

Solution:

The equation is $\frac{dy}{dx} + \left(\frac{x\cos x + \sin x}{x\sin x}\right)y = \frac{1}{x}$

Integrating factor I.F. $= \mathrm{e}^{\int \frac{x\cos x + \sin x}{x\sin x}dx} = e^{\log(x\sin x)} = x\sin x$

∴ The solution is

$y(x\sin x) = \int \frac{1}{x}(x\sin x)dx + c$

$xy\sin x = -\cos x + c$ **when**

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$x = 0, y = 0 \Rightarrow c = \cos 0 = 1$

$\therefore$ **The particular solution is** $xy \sin x = 1 - \cos x$

Q.10 Answer:

$\frac{3y}{x+1} = e^{3x} + c$

Solution:

The given equation is $\frac{dy}{dx} - \frac{y}{x+1} = e^{3x}(x+1)$

I.F. $= e^{\int -\frac{1}{x+1} dx} = e^{-\log(x+1)} = \frac{1}{x+1}$

The solution is $y\left(\frac{1}{x+1}\right) = \int e^{3x}(x+1) \cdot \frac{1}{x+1} dx + a$

$\Rightarrow \frac{y}{x+1} = \int e^{3x} dx + a = \frac{e^{3x}}{3} + a$

$\Rightarrow \frac{3y}{x+1} = e^{3x} + c, c = 3a$

Q.11 Answer:

$y\left(1 + x^2\right) = c + \tan^{-1} x$

Solution:

Given differential equation 1 s

$\frac{dy}{dx} + \frac{2yx}{1+x^2} = \frac{1}{(1+x^2)^2}$ **which is a linear.**

Differential equation of the form: $\frac{dy}{dx} + \mathrm{P}y = \mathrm{Q}$ **On comparing, we have** $\mathrm{P} = \frac{2x}{1+x^2}$ **and** $Q = \frac{1}{(1+x^2)^2}$

I.F $= e^{\int 2x/1+x^2 dx} = e^{\log\left(1+x^2\right)} = \left(1 + x^2\right)$

$\therefore$ **Solution is** $y\left(1 + x^2\right)$

$= \int \frac{1}{(1+x^2)^2}\left(1 + x^2\right) dx + c$

$\Rightarrow y\left(1 + x^2\right) = \int \frac{1}{(1+x^2)} dx + c$

$\Rightarrow y\left(1 + x^2\right) = \tan^{-1} x + c$

Q.12 Answer:

$\sin^2 x$

Solution:

Given differential equation is $\sin x \frac{dy}{dx} + 2y \cos x = 1; \Rightarrow \frac{dy}{dx} + 2y\frac{\cos x}{\sin x} = \frac{1}{\sin x}$

$\Rightarrow \frac{dy}{dx} + (2 \cot x)y = \operatorname{cosec} x$

$I.F = e^{\int 2 \cot x dx} = e^{\int_2 \left(\frac{\cos x}{\sin x}\right) dx} = e^{2 \log \sin x}$

$= e^{\log_e (\sin x)^2} = (\sin x)^2$

Q.13 Answer:

$\sin y = (1 + x)\left(e^x + c\right)$

Solution:

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Divide the equation by sec $y \cos y \frac{dy}{dx} - \frac{\sin y}{1+x} = (1+x)e^x$
Put $\sin y = z \Rightarrow \cos y \frac{dy}{dx} = \frac{dz}{dx}$ **then**
$\frac{dz}{dx} - \left(\frac{1}{1+x}\right) z = (1+x)e^x$
I.F. $= e^{-\int \frac{1}{1+x} dx} = e^{-\log(1+x)} = \frac{1}{1+x}$
The solution is $z\left(\frac{1}{1+x}\right) = e^x + c \Rightarrow \sin y = (1+x)\left(e^x + c\right)$

Q.14 Answer:

$x = \log y \left(cx^2 + \frac{1}{2}\right)$

Solution:

Divide the equation by $y(\log y)^2$
$\frac{1}{y(\log y)^2} \frac{dy}{dx} + \frac{1}{\log y} \cdot \frac{1}{x} = \frac{1}{x^2}$
Put $\frac{1}{\log y} = z \Rightarrow \frac{-1}{y(\log y)^2} \frac{dy}{dx} = \frac{dz}{dx}$
Thus, we get, $-\frac{dz}{dx} + \frac{1}{x} \cdot z = \frac{1}{x^2}$, linear in z
$\Rightarrow \frac{dz}{dx} + \left(-\frac{1}{x}\right) z = -\frac{1}{x^2}$
I.F. $= e^{-\int \frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$
The solution is, $z\left(\frac{1}{x}\right) = \int \frac{-1}{x^2}\left(\frac{1}{x}\right) dx + c$
$\Rightarrow \frac{1}{\log y}\left(\frac{1}{x}\right) = \frac{-x^{-2}}{-2} + c \Rightarrow x = \log y \left(cx^2 + \frac{1}{2}\right)$

Q.15 Answer:

$x^x = e^{ye^y}$

Solution:

$(1+\log x)\frac{dx}{dy} - x \log x = e^y$
putting $x \log x = t \Rightarrow (1+\log x)dx = dt$
$\therefore \quad \frac{\mathrm{dt}}{\mathrm{dy}} - \mathrm{t} = \mathrm{e}^{\mathrm{y}} \quad$ **Now, I.F.** $= \mathrm{e}^{\int -1 \mathrm{dy}} = \mathrm{e}^{-\mathrm{y}}$
$\Rightarrow \quad te^{-y} = \int e^{-y} e^y dy + C$
$\Rightarrow \quad \mathrm{t} = \mathrm{Ce}^{\mathrm{y}} + \mathrm{ye}^{\mathrm{y}} \Rightarrow \mathrm{x} \log \mathrm{x} = (\mathrm{C} + \mathrm{y})\mathrm{e}^{\mathrm{y}}$
Since, $y(1) = 0$**, then** $0 = (\mathrm{C}+0)1 \Rightarrow \mathrm{C} = 0$
$\therefore ye^y = x \log x \Rightarrow x^x = e^{ye^y}$

Q.16 Answer:

$f(x) = y(x+c)$

Solution:

We have $\frac{dy}{dx} = \frac{f'(x)}{f(x)} y - \frac{y^2}{f(x)}$
$\Rightarrow \frac{dy}{dx} - \frac{f'(x)}{f(x)} y = -\frac{y^2}{f(x)}$
Divide by y^2 : $y^{-2}\frac{dy}{dx} - y^{-1}\frac{f'(x)}{f(x)} = -\frac{1}{f(x)}$
Put $y^{-1} = z \Rightarrow -y^{-2}\frac{dy}{dx} = \frac{dz}{dx}$

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$-\frac{dz}{dx}-\frac{f'(x)}{f(x)}(z)=-\frac{1}{f(x)}$

$\Rightarrow \frac{dz}{dx}+\frac{f'(x)}{f(x)}z=\frac{1}{f(x)}$

I.F. $=e^{\int \frac{f'(x)}{f(x)}dx}=e^{\log f(x)}=f(x)$

∴ The solution is $z(f(x))=\int \frac{1}{f(x)}(f(x))dx+c$

$\Rightarrow y^{-1}(f(x))=x+c \Rightarrow f(x)=y(x+c)$

Q.17 Answer:

$y=\{\phi(x)-1\}+ce^{-\phi(x)}$

Solution:

We have, $dy+\{y\phi'(x)-\phi(x)\phi'(x)\}dx=0$

$\Rightarrow \quad \frac{dy}{dx}+\phi'(x)\cdot y=\phi(x)\phi'(x)$

This is a linear differential equation with $\mathrm{IF}=e^{\int \phi'(x)dx}=\mathrm{e}^{\phi(x)}$

Multiplying Eq. (i) by $\phi(x)$ **and integrating, we get** $ye^{\phi(x)}=\int \phi(x)\phi'(x)e^{\phi(x)}dx$

$\Rightarrow \quad ye^{\phi(x)}=\int e^{\phi(x)}\phi(x)\phi'(x)dx$

$\Rightarrow ye^{\phi(x)}=\int \phi(x)e^{\phi(x)}\phi'(x)dx$

$\Rightarrow ye^{\phi(x)} \quad =\phi(x)e^{\phi(x)}-\int \phi'(x)e^{\phi(x)}dx$

$\Rightarrow \quad ye^{\phi(x)}=\phi(x)e^{\phi(x)}-e^{\phi(x)}+c$

$\Rightarrow \quad y=\{\phi(x)-1\}+ce^{-\phi(x)}$

Q.18 Answer:

$4xy=y^4+c$

Solution:

$\frac{dx}{dy}+\frac{x}{y}-y^2=0; \frac{dx}{dy}+\frac{x}{y}=y^2$

This is a linear differential equation of the form $\frac{\mathrm{dx}}{\mathrm{dy}}+\mathrm{P_1x}=\mathrm{Q_1}$**; Here,** $\mathrm{P}=\frac{1}{\mathrm{y}}$ **and** $\mathrm{Q}=\mathrm{y}^2$

$\therefore \quad$ **I.F.** $=\mathrm{e}^{\int \mathrm{Pdy}}=\mathrm{e}^{\int \frac{1}{\mathrm{y}}\mathrm{dy}}=\mathrm{e}^{\log \mathrm{y}}=\mathrm{y}$

So, required solution is $x\cdot y=\int y^2\cdot ydy+c; xy=\int y^3dy+c$

$xy=\frac{y^4}{4}+c; 4xy=y^4+c$

Q.19 Answer:

$y-x+\ln(x+y)=c$

Solution:

Differential equation is $(x+y)(dx-dy)=dx+dy$

dividing by dx **on both the sides** $(x+y)\left(1-\frac{dy}{dx}\right)=1+\frac{dy}{dx}$

Putting $\mathrm{x+y=v}$

$1+\frac{\mathrm{dy}}{\mathrm{dx}}=\frac{\mathrm{dv}}{\mathrm{dx}}$ **and** $\frac{\mathrm{dy}}{\mathrm{dx}}=\frac{\mathrm{dv}}{\mathrm{dx}}-1$

The equation changes to $v\left\{1-\left(\frac{dv}{dx}-1\right)\right\}=\frac{dv}{dx}; v\left(2-\frac{dv}{dx}\right)=\frac{dv}{dx}$

$2v-v\frac{dv}{dx}=\frac{dv}{dx}; 2v=(1+v)\frac{dv}{dx}$

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$\left(\frac{1+v}{v}\right) dv = 2dx$ **or** $\left(\frac{1}{v}+1\right) dv = 2dx$

Integrating on both the sides,

$\int \frac{dv}{v} + \int dv = 2\int dx + c$

$\log v + v = 2x + c$

Putting $\mathrm{v} = \mathrm{x} + \mathrm{y}$

$\log(x+y) + x + y = 2x + c$

or, $\log(x+y) + y - x = c$

or,$y - x + \log(x+y) = c$

Q.20 Answer:

ellipses and hyperbolas

Solution:

The given equation is linear in y **and can be written as**

$\frac{dy}{dx} + \frac{x}{1-x^2} y = \frac{ax}{1-x^2}$

Its integrating factor is

$e^{\int \frac{x}{1-x^2}^x} = e^{-(12)\log\left(1-x^2\right)} = \frac{1}{\sqrt{1-x^2}}$

if $-1 < x < 1$ **and if** $x^2 > 1$

then I.F. $= \frac{1}{\sqrt{x^2-1}}$

$\frac{d}{dx}\left(y\frac{1}{\sqrt{1-x^2}}\right) = \frac{\alpha x}{(1-x^2)^{3/2}} = -\frac{1}{2}a\frac{-2}{(1-x^2)^{3/2}}$

$\Rightarrow y\frac{1}{\sqrt{1-x^2}} = \frac{a}{\sqrt{1-x^2}} + C$

$\Rightarrow y = a + C\sqrt{1-x^2}$

$\Rightarrow (y-a)^2 = C^Q\left(1-x^2\right)$

$\Rightarrow (y-a)^2 + C^2x^2 = C^2$

Thus if $-1 < x < 1$ **the given equation represents an ellipse . If** $x^2 > 1$ **then the sohtion is of the form** $-(y-a)^2 - C^2x^2 = C^2$ **which represents a hyperbola.**

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